

ENVIRONMENTAL ACCOUNTABILITY INITIATIVE

Institutional Regulatory Examination & Pilot Reference | Version 3.0

From clean-air commitments to admissible evidence, applicable authority, established standing, and verified consequence.

10 October 2026 | Founder: Gregory Don Butler

RELEASE STATUS: Proposed architecture; pre-examination; baseline for external technical review. Not regulator-approved, accredited, certified, legally binding, or field-validated.

Purpose: Invite a qualified institution to identify one concrete environmental obligation or voluntary commitment and independently test whether EIG and AIR improve the traceability of evidence and consequential decisions.

Document control and freeze statement

Control	Declaration
Reference	TA14 Environmental Accountability Initiative / v3.0
Baseline	Architecture and proposed pilot methodology frozen for invitation; no operational validation
Change from v2.0	Worked decision trace; emergency-action clarification; cross-architecture integration; examiner reproducibility; pilot governance
Review authority	Institutional reviewer determines accept for examination / revise / hold / decline
Non-claims	No statutory mandate for AIR; no automatic courtroom admissibility; no general EU AI Act high-risk classification; no universal public-health-building designation

This release is a proposed examination baseline, not a statement of regulatory equivalence. Every application requires a locally validated applicability register and a named competent counterpart.

1 | Executive proposition

Public pledges, binding laws, contracts and certification schemes establish different obligations. An environmental number, however accurately observed, does not itself establish that a promise was fulfilled, a legal duty was met, an intervention was permitted, or a human health outcome occurred.

Environmental Integrity Governance (EIG) examines evidence, authority, standing, constraints, decision and consequence. **Atmospheric Integrity Records (AIR)** preserve original observations, measurement context, provenance, correction and challenge history. Both are overlays; neither is the sensor, building automation system, digital twin, regulator or legal authority.

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Invariant chain: REALITY → RECORD → CONTINUITY → ADMISSIBILITY → BINDING → COMMIT → EXECUTION → OUTCOME.

2 | Regulatory applicability: exact duties, not analogy

A candidate instrument is not a determination of applicability. For every pilot, an independent legal/technical counterpart must establish the precise instrument, operative provision, relevant date, implementation measure, building category, dutyholder, evidence duty, regulator and review process.

Candidate source	Question to test	Boundary
EU Directive (EU) 2024/1275 (EPBD)	Which transposed national provisions create an indoor environmental quality, monitoring or controls duty for the selected building?	Implementation and deadlines vary by state, provision and building.

Candidate source	Question to test	Boundary
EU Regulation (EU) 2024/1689 (AI Act)	Does the specific AI use meet the applicable definition and classification; which obligations and dates apply?	Not all HVAC AI or building automation is high-risk.
England Building Safety Act 2022 and regulations	Is this a covered higher-risk building and prescribed golden-thread information?	Not a general mandate for all UK buildings or all IAQ records.
World Health Network Clean Air Pledge	Which optional commitment was selected; what metrics, dates and verification terms govern it?	Voluntary pledge is not statutory enforcement authority.

Jurisdiction register — required fields

Jurisdiction | official source URL | article/section | consolidated version and effective date | national implementing provision | building/system scope | dutyholder | measurable obligation | record and retention rule | authorized reviewer | enforcement or contractual consequence | unresolved legal question | reviewer sign-off.

The BUILD UP characterization of buildings as public-health infrastructure is policy framing, not a blanket statutory designation. Claims of compliance, health benefit, or legal evidentiary status must be examined separately.

3 | AIR minimum record and evidence quality

Family	Candidate required content
Identity	Record ID; building/zone; event and observation windows; custodian; purpose
Observation	Unmodified raw value, unit, time zone, interval, instrument ID, placement and method
Quality	Calibration trace, uncertainty, drift, maintenance, detection limits, missing data
Context	System operating mode, filtration, pressure, outdoor conditions, lawful occupancy context
Integrity	Source provenance, checksums or equivalent, version and access/custody/export logs
Adjustment	Separate corrected/normalized value, reason, author, method, uncertainty and original reference
Challenge	Competing evidence, dispute ID, reviewer, status, dissent preserved
Consequence	Decision ID, authority citation, permitted action, execution trace, post-action verification

Synthetic AIR event — explicitly not field evidence

Attribute	Illustrative entry
Record	AIR-DEMO-SCHOOL-001 / zone classroom A / 10 Oct 2026 09:00 EDT
Raw measurement	PM2.5 42 micrograms/m3; sensor S-17; method and averaging window not established
Quality	Calibration UNVERIFIED; measurement uncertainty UNKNOWN
Context	Supply fan ON; possible smoke intrusion, cause not yet established
Evidence disposition	HOLD for definitive quantitative or compliance conclusion; corroboration requested
Preservation	Raw observation retained unchanged; no retrospective correction
Safety branch	Separate existing school emergency authority and precautionary protocols must still be evaluated promptly

4 | EIG examination and safety exception boundary

An EIG disposition applies to a particular proposed consequence. It must not be misread as a blanket prohibition on all action when one measurement is uncertain. If independent emergency duties, precautionary operating procedures, or competent human emergency authority apply, they must be identified and examined on their own basis.

Gate	Required question
Evidence	Is this evidence fit for this particular claim and decision?
Applicable authority	Which law, approved procedure, contract or delegated operational instruction governs?
Standing	Who may act, where, on what asset, within which time and delegation?
Safety	Are emergency protocols, interlocks, life-safety constraints and escalation routes satisfied?
Commit	Was an attributable, bounded, versioned authorization recorded before the governed action?
Verify	Can an independent party confirm the command and bounded environmental result?

ALLOW = declared criteria met for a bounded action; **HOLD** = insufficient basis for that proposed action or claim; **DENY** = valid prohibition or constraint; **ESCALATE** = competent authority must decide. These are proposed architectural labels, not governmental rulings.

Emergency clarification

Example: a school sensor has uncertain calibration while smoke is suspected. HOLD may apply to a claim that a precise threshold was exceeded. It does not automatically prevent a principal or operator from invoking a separately authorized precautionary shelter-in-place, filtration, evacuation or emergency procedure. The emergency authority, trigger, safety trade-offs, action and later review must each be recorded. No design should delay urgent legally required protection while awaiting perfect evidence.

5 | Integration with the TA14 governance architecture

Component	Role in proposed environmental use
AIR	Evidence continuity: preserves what was observed and what remains uncertain
EIG	Environmental decision governance: evaluates commitments, evidence and accountable consequences
AEA	Admissible execution gate: proposed consequence may execute only when conditions are established
AFA	Federation: authority context and evidence may cross organizational boundaries; local execution authority remains necessary
EABA	Execution boundary: constrains which entity or system can commit an action
Authority Passport (AVP)	Candidate machine-readable representation of scope, validity and authority claims; not itself legal authorization

This mapping is conceptual, not proof that these components are implemented together or independently validated. A pilot must identify exact versions, interface contracts, trust anchors, failure behavior, and the organization that owns each decision.

6 | Worked trace A: school smoke event

Anchor	Synthetic event and examinable artifact
REALITY	Potential smoke intrusion; report from school operations; event opened
RECORD	AIR-DEMO-SCHOOL-001; raw PM2.5 reading 42 micrograms/m3; unknown calibration
CONTINUITY	Original reading and missing calibration metadata preserved; integrity manifest created
ADMISSIBILITY	HOLD on numeric compliance claim; request calibrated corroboration and outdoor comparison
BINDING	School emergency procedure and operator authority independently checked; no inferred permission from alert
COMMIT	Authorized operator records precautionary filtration action under valid procedure, limits and review time
EXECUTION	Permitted filtration action only; do not change pressure or smoke-control modes without appropriate engineering authorization
OUTCOME	Post-action observations compared under declared method; no automatic causal or clinical health claim

Illustrative decision record

Decision EIG-DEMO-001 | Proposed claim: "classroom breached an enforceable threshold" | HOLD because method/calibration insufficient. Separate decision EIG-DEMO-002 | Proposed action: "activate approved precautionary filtration protocol" | ALLOW only if the school procedure, actor, equipment constraints and expiration are independently established. Neither disposition proves the other.

7 | Worked trace B: unauthorized AI control

An AI agent predicts that increasing outside air will reduce an indoor pollutant. The model has network access to a building control interface but no verified delegation to override smoke-control or emergency modes. Prediction is inference, not observation; authentication is not standing.

Test	Expected trace
Authority check	No applicable delegation for override; authority context fails local validation
Disposition	DENY proposed prohibited override; ESCALATE possible alternate protective action
Execution gate	No governed command issued; attempted request and denial logged
Independent verification	Examiner compares request, authority state, gateway decision and actuator/event logs
Counterfactual	A separately valid, bounded and safe delegated action can be ALLOW; system must not always deny

The pilot must test both permissive and adverse branches, including a forged or expired credential, a disconnected gateway, a changed emergency mode and a contradictory instruction. Failure to block an unauthorized command is a critical failure.

8 | Independent examination: reproducible method

An examiner independent of the implementer must receive a frozen test plan, source manifest, record exports, decision rules, instrument-quality metadata, local authority references, command traces and expected outcomes. The examiner should be able to reproduce the disposition without relying on a TA14-authored narrative alone.

Minimum test sequence

- Freeze jurisdiction, duty or pledge, building type, systems, baseline, evidence window and exclusions.
- Reconstruct one raw observation from the source instrument and verify time, unit, provenance and quality metadata.
- Recalculate one adjusted value without overwriting the original; preserve alternative explanations.
- Reproduce an ALLOW, HOLD, DENY and ESCALATE decision from explicit evidence and authority references.
- Attempt unauthorized execution and independently verify absence of actuation across all relevant pathways.
- Reproduce a permitted execution and verify actual command, asset response and bounded outcome.
- Test emergency authority separately from an uncertain environmental claim; verify no dangerous procedural deadlock.
- Report disagreements, missed records, false conclusions, examiner hours, cost and unresolved limitations.

Category	Acceptance evidence	Failure / hold
Integrity	Raw originals and verifiable custody/export	Overwritten source or irrecoverable custody
Quality	Uncertainty and calibration limitations visible	Unqualified reading treated as established truth
Authority	Locally verified actor, action, scope and time	Expired or missing authority permits execution
Safety	Emergency paths evaluated independently	Governance blocks mandatory emergency response or bypasses interlocks
Outcome	Independent bounded reproduction	Unsupported causal, health or compliance conclusion
Operations	Workload, cost and exceptions reported	Material burden or false conclusions hidden

A negative finding is a valid examination outcome. Reviewer disagreement is preserved, not edited away. No acceptance threshold should be retrofitted after results are observed.

9 | Security, privacy, retention and legal process

- Collect only lawful, necessary personal or occupancy data; identify controller, custodian, purpose, legal basis and access rights.
- Use tamper-evident preservation, role-based access, least privilege, key management, secure transmission and time synchronization.
- Define retention, legal hold, archival and deletion policies according to actual jurisdictional requirements; never promise indefinite retention by default.
- Preserve correction history, contrary evidence and dispute resolution; provide reproducible export without granting the recipient execution authority.
- Do not equate a TA14 evidence label with court admissibility, accredited certification or a regulatory enforcement determination.

10 | Pilot charter and institutional offer

Stage	Proposed deliverable	Decision
A. Introductory technical briefing	30–45 minute scoping conversation; identify one obligation and named technical counterpart	Proceed / decline
B. Bounded examination	Signed scope, baseline, legal applicability worksheet, test artifacts, confidentiality, examiner criteria	Approve / revise / hold
C. Funded pilot	Instrument/source integration, authority mapping, adversarial tests, operational cost and independent report	Accept bounded findings / reject
D. Standards or procurement discussion	Only with sufficient evidence, authorization and partner interest	Separate institutional determination

Proposed roles and commercial boundaries

Institutional sponsor selects a legitimate use case and provides a qualified counterpart. Dutyholder confirms actual legal and operational responsibilities. Implementer prepares instrument and systems interfaces. Independent examiner controls evaluation criteria and records findings. TA14 supplies proposed EIG/AIR methods and architecture documentation, without presuming official authority. Budget, schedule, licensing, IP terms and procurement must be negotiated rather than assumed.

Pilot go / no-go

GO only after named counterpart, legal applicability review, written scope, instrument access, independent examiner, emergency procedures, security/privacy arrangements and funding are agreed. NO-GO if the partner requests a blanket compliance certification, cannot identify a competent dutyholder, lacks lawful access to records, or cannot permit meaningful adverse-path testing.

11 | Practitioner objections retained as test requirements

Liability: architecture cannot compel unwilling owners absent applicable law, contract or policy. **Enforcement capacity:** measure actual reviewer workload and cost. **Sensor accuracy:** preserve original readings with calibration, uncertainty, corrections and dissent. **Paperwork without consequence:** verify that an unauthorized act is actually stopped. **Jurisdictional variation:** preserve the invariant examination question while binding local duties and limits.

12 | Ten-image public showroom alignment

01 The Promise | 02 Human Infrastructure | 03 Regulatory Obligation | 04 Measurement Limits | 05 AIR | 06 Authority Boundary | 07 Intervention | 08 Verified Outcome | 09 Independent Examination | 10 Institutional Invitation.

The showroom must label all synthetic data as illustrative, display PRE-EXAMINATION / NO REGULATORY ENDORSEMENT, link to this reference, and offer a clearly scoped institutional contact pathway. Teaching images are not measurement evidence.

13 | Sources and verification register

- R1. Directive (EU) 2024/1275 (EPBD): <https://eur-lex.europa.eu/eli/dir/2024/1275/oj>
- R2. Regulation (EU) 2024/1689 (AI Act): <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>
- R3. Building Safety Act 2022: <https://www.legislation.gov.uk/ukpga/2022/30/contents>
- R4. World Health Network Clean Air Pledge: <https://whn.global/clean-air-pledge/>
- R5. European Commission BUILD UP: <https://build-up.ec.europa.eu/>

Verification status: These are source locators, not independently verified section-level legal opinions. Before external reliance, confirm official consolidated versions, exact provisions, national transposition, applicability dates, specific pledge elections, instrument suitability and any relevant administrative guidance.

Institutional invitation

TA14 Authority Governance Institution invites a qualified regulator, public-health institution, standards organization or institutional dutyholder to identify one precisely bounded environmental accountability requirement and examine whether EIG and AIR can improve evidence continuity, authorized decision-making and independent verification.

No admissible evidence. No admissible execution. | Capability is not authority. | Observed is not inferred. Inferred is not authorized.